

Aron Rez D. Arboleda

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EDUCATION

Tarlac State University | Tarlac, Philippines Graduation Date: July 2026
Bachelor of Science in Computer Science Final GWA: 1.24 (96.12%, Magna Cum Laude & Class Valedictorian)

Technical Skills: TypeScript | JavaScript | Python | Java | C++ | PHP | React | HTML | CSS | TailwindCSS | Node.js | FastAPI | Electron | Java Swing | Visual Basic | React Native + Expo | MySQL | Firebase | MongoDB

WORK EXPERIENCE

Freelance Software Developer | Tarlac, Philippines Mar. 2025 – May 2025 & Mar. 2026 – May 2026

- Collaborated with a UI/UX designer and 10 student engineers to develop the Draft2Dimen application, a tool for civil engineers to accurately estimate the sizes of structural components of a building.
- Implemented CRUD functions and local saving using Electron, React, Typescript, and TailwindCSS to further create user-friendly interfaces and navigations.
- Built a functional Geographical Information System (GIS) called Liquefact by collaborating with clients, it is an application that predicts borehole information in Tarlac, Philippines.
- Integrated and trained a customized neural network with a FastAPI backend, facilitated the whole machine learning process to build and deploy the model.

Trackerteer Web Developer Corporation | Clark, Philippines June 2025 - July 2025
Project Assistant Internship

- Developed the Inventeer web application, a tool that automates the process of generating tables for product inventory and RFID management.
- Helped the media team by producing a total of 7 user guides averaging 5 mins in length. Facilitated the whole process of scripting, recording, video editing, and audio work.
- Maintained and consistently updated the inventory tables by processing client requests and generating SGTIN-96 codes for RFID tracking.

PROJECTS

U-Heal | Bachelor's Thesis | React Native, PyTorch, FastAPI, Firebase Jan. 2026 - Apr. 2026

- Acted as the main leader and sole developer of our Bachelor's Thesis, a full-stack mobile eHealth system for remote chronic wound monitoring. U-Net architecture was integrated to enhance the system's functionality and address the inter-rater variability from manual tissue estimation.
- Architected a full-stack eHealth platform supporting wound documentation, real-time messaging, video consultations, appointment scheduling, AI-powered wound analysis, and longitudinal healing tracking for clinicians and patients.

Draft2Dimen | Electron, React PDF, MathJS | Desktop Application

- Version 1 Mar. 2025 – May 2025
 - Built an application for civil engineers to automate the tedious process of calculating the appropriate sizes of structural components of a building.
 - Implemented the "Newton's iteration" formula and MathJS library to reduce the delay of mathematical computation by 70%. A total of 4 computation modules were implemented with PDF export features using React PDF library to ensure proper usage documentation.
- Version 2 Apr. 2026 – May 2026
 - Added 2 steel computation modules, cost computation, and utilized localStorage JS functionality for the saved runs CRUD table to provide a persistent storage layer for the application.

Liquefact | React, TypeScript, PyTorch, FastAPI | Web Application May 2026

- Developed a Geographical Information System (GIS) to help predict boreholes around the Tarlac province in the Philippines. Integrated a multi-task neural network with 6 regression and 2 categorical outputs; and implemented multi-threading technique using Web Workers to compute and render all the borehole points 4x the speed.

Other Projects | React, PHP, MySQL | Full-Stack Web Applications

- Liwanag at Dunong Website** Nov. 2024 - Jan. 2025
Built a landing page with an admin dashboard to facilitate volunteer submissions, implement access control, and monitor website analytics.
- REBYU: Gamified Flashcards** Nov. 2024 - Dec. 2024
Engineered a Quizlet-like web application with added gamification by applying pixel art graphics to engage users in studying virtual flashcards.

ADDITIONAL

Certificates: RAITE Hackathon Regional Competition certificate of participation, Nov. 2024 | Cisco "CCNAV7: Switching, Routing, and Wireless Essentials", June 2024 | Cisco "CCNAV7: Introduction to Networks", Jan 2024.

Involvement: Liwanag at Dunong (NGO)—developed the organization's website and helped with volunteer work for aeta indigenous people, Jul. 2024 - Jan. 2025 | Became a member of Programmers' Den (University Organization), Oct. 2022 - Dec 2022. | College Chess Team, 3rd Place both on 2024 and 2025 Intramurals.